2.59 Variable Acceleration: Vectors

Worked Solutions

1 a)

$\mathbf{v} = \frac{d\mathbf{s}}{dt} = (12t^2)\mathbf{i} + \mathbf{j}$

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (24t)\mathbf{i}$

when $t = 0.5$, $\mathbf{a} = (24 \times 0.5)\mathbf{i} = 12\mathbf{i}$

$|\mathbf{a}| = 12 \text{ ms}^{-2}$

1 b)

$\mathbf{v} = \frac{d\mathbf{s}}{dt} = (9t^2)\mathbf{i} + (-4t^3)\mathbf{j}$

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (18t)\mathbf{i} + (-12t^2)\mathbf{j}$

when $t = 2$, $\mathbf{a} = (18 \times 2)\mathbf{i} + (-12 \times 2^2)\mathbf{j}$

$= (36)\mathbf{i} + (-48)\mathbf{j}$

$|\mathbf{a}| = \sqrt{36^2 + 48^2} = 60 \text{ ms}^{-2}$

1 c)

$\mathbf{v} = \frac{d\mathbf{s}}{dt} = (e^{0.5t})\mathbf{i} + (2t)\mathbf{j}$

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (0.5e^{0.5t})\mathbf{i} + 2\mathbf{j}$

at $t = 0.1$, $\mathbf{a} = (0.5e^{0.5\times0.1})\mathbf{i} + 2\mathbf{j}$

$= (0.5e^{0.05})\mathbf{i} + 2\mathbf{j}$

$|\mathbf{a}| = \sqrt{(0.5e^{0.05})^2 + 2^2} = 2.07 \text{ ms}^{-2}$ (3 SF)

2 a)

$\mathbf{v} = \int \mathbf{a}\, dt = (2t^2 - 5t + c_1)\mathbf{i} + (3t^3 + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{u} = 3\mathbf{i} - \mathbf{j}$

$0 - 0 + c_1 = 3 \Rightarrow c_1 = 3$

$0 + c_2 = -1 \Rightarrow c_2 = -1$

$\mathbf{v} = (2t^2 - 5t + 3)\mathbf{i} + (3t^3 - 1)\mathbf{j}$

when $t = 5$

$\mathbf{v} = (2 \times 5^2 - 5 \times 5 + 3)\mathbf{i} + (3 \times 5^3 - 1)\mathbf{j}$

$= 28\mathbf{i} + 374\mathbf{j}$

$|\mathbf{v}| = \sqrt{28^2 + 374^2} = 375 \text{ ms}^{-1}$ (3 SF)

2 b)

$\mathbf{v} = \int \mathbf{a}\, dt = (6t^2 + 2t + c_1)\mathbf{i} + (-18t + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{u} = -2\mathbf{i} + 5\mathbf{j}$

$0 + 0 + c_1 = -2 \Rightarrow c_1 = -2$

$0 + c_2 = 5 \Rightarrow c_2 = 5$

$\mathbf{v} = (6t^2 + 2t - 2)\mathbf{i} + (-18t + 5)\mathbf{j}$

when $t = 0.5$

$\mathbf{v} = (6 \times 0.5^2 + 2 \times 0.5 - 2)\mathbf{i} + (-18 \times 0.5 + 5)\mathbf{j}$

$= 0.5\mathbf{i} - 4\mathbf{j}$

$|\mathbf{v}| = \sqrt{0.5^2 + 4^2} = 4.03 \text{ ms}^{-1}$ (3 SF)

3 a)

$\mathbf{s} = \int \mathbf{v}\, dt = (\frac{1}{2}t^2 - 5t + c_1)\mathbf{i} + (3t + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{s} = -4\mathbf{i} + 10\mathbf{j}$

$0 - 0 + c_1 = -4 \Rightarrow c_1 = -4$

$0 + c_2 = 10 \Rightarrow c_2 = 10$

$\mathbf{s} = (\frac{1}{2}t^2 - 5t - 4)\mathbf{i} + (3t + 10)\mathbf{j}$

when $t = 0.4$

$\mathbf{s} = (\frac{1}{2} \times 0.4^2 - 5 \times 0.4 - 4)\mathbf{i} + (3 \times 0.4 + 10)\mathbf{j}$

$= (-5.92)\mathbf{i} + (11.2)\mathbf{j}$

$|\mathbf{s}| = \sqrt{5.92^2 + 11.2^2} = 12.7$ m (3 SF)

3 b)

$\mathbf{s} = \int \mathbf{v}\, dt = (\frac{3}{2}t^2 + c_1)\mathbf{i} + (-20e^{-0.1t} + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{s} = 2\mathbf{i} + 0.5\mathbf{j}$

$0 + c_1 = 2 \Rightarrow c_1 = 2$

$-20 + c_2 = 0.5 \Rightarrow c_2 = 20.5$

$\mathbf{s} = (\frac{3}{2}t^2 + 2)\mathbf{i} + (-20e^{-0.1t} + 20.5)\mathbf{j}$

when $t = 0.1$

$\mathbf{s} = (\frac{3}{2} \times 0.1^2 + 2)\mathbf{i} + (-20e^{-0.1\times0.1} + 20.5)\mathbf{j}$

$= (2.015)\mathbf{i} + (-20e^{-0.01} + 20.5)\mathbf{j}$

$|\mathbf{s}| = \sqrt{2.015^2 + (-20e^{-0.01} + 20.5)^2}$

$= 2.14$ m (3 SF)

4 a)

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 5\mathbf{i} + (2t - 2)\mathbf{j}$

$\mathbf{F} = m\mathbf{a} = (5\mathbf{i} + (2t - 2)\mathbf{j}) \times 1.2$

$= 6\mathbf{i} + (2.4t - 2.4)\mathbf{j}$

When $t = 5$, $\mathbf{F} = 6\mathbf{i} + (2.4 \times 5 - 2.4)\mathbf{j} = 6\mathbf{i} + 9.6\mathbf{j}$

$|\mathbf{F}| = \sqrt{6^2 + 9.6^2} = 11.3$ N (3 SF)

4 b)

$\mathbf{s} = \int \mathbf{v}\, dt = (\frac{5}{2}t^2 + 2t + c_1)\mathbf{i} + (\frac{1}{3}t^3 - t^2 + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{s} = 2\mathbf{i} + 3\mathbf{j}$

$0 + 0 + c_1 = 2 \Rightarrow c_1 = 2$

$0 - 0 + c_2 = 3 \Rightarrow c_2 = 3$

$\mathbf{s} = (\frac{5}{2}t^2 + 2t + 2)\mathbf{i} + (\frac{1}{3}t^3 - t^2 + 3)\mathbf{j}$

when $t = 5$

$\mathbf{s} = (\frac{5}{2} \times 5^2 + 2 \times 5 + 2)\mathbf{i} + (\frac{1}{3} \times 5^3 - 5^2 + 3)\mathbf{j}$

$= 74.5\mathbf{i} + \frac{59}{3}\mathbf{j}$

$|\mathbf{s}| = \sqrt{74.5^2 + (\frac{59}{3})^2} = 77.1$ m (3 SF)

5)

$\mathbf{s} = \int_1^5 \mathbf{v}\, dt = \int_1^5 [(6t^2)\mathbf{i} + (4t - 1)\mathbf{j}]dt$

$= [(2t^3)\mathbf{i} + (2t^2 - t)\mathbf{j}]_1^5$

$= [(2 \times 5^3)\mathbf{i} + (2 \times 5^2 - 5)\mathbf{j}] - [(2 \times 1^3)\mathbf{i} + (2 \times 1^2 - 1)\mathbf{j}]$

$= (250\mathbf{i} + 45\mathbf{j}) - (2\mathbf{i} + \mathbf{j}) = 248\mathbf{i} + 44\mathbf{j}$ metres

$|\mathbf{s}| = \sqrt{248^2 + 44^2} = 252$ m (3 SF)

6)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (4t - 1)\mathbf{i} + 2\mathbf{j}$

$|\mathbf{r}| = \sqrt{(4t - 1)^2 + 2^2} = \sqrt{16t^2 - 8t + 1 + 4}$

$= \sqrt{16t^2 - 8t + 5}$

$2 = \sqrt{16t^2 - 8t + 5}$

$4 = 16t^2 - 8t + 5 \Rightarrow 16t^2 - 8t + 1 = 0$

$t = 0.25$ s

7)

$\mathbf{a} = \int_0^1 \mathbf{v}\, dt = \int_0^1 [(2t + 9)\mathbf{i} + (e^{-0.1t})\mathbf{j}]dt$

$= [(t^2 + 9t)\mathbf{i} + (-10e^{-0.1t})\mathbf{j}]_0^1$

$= [(1 + 9)\mathbf{i} + (-10e^{-0.1})\mathbf{j}] - [(0 + 0)\mathbf{i} + (-10)\mathbf{j}]$

$= (10\mathbf{i} - 10e^{-0.1}\mathbf{j}) + 10\mathbf{j}$

$= 10\mathbf{i} + (10 - 10e^{-0.1})\mathbf{j}$ metres

8 a)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (2t + 1)\mathbf{i} + 5\mathbf{j}$

when $t = 4$, $\mathbf{v} = (2 \times 4 + 1)\mathbf{i} + 5\mathbf{j} = 9\mathbf{i} + 5\mathbf{j}$

$|\mathbf{v}| = \sqrt{9^2 + 5^2} = 10.3$ ms^{-1} (3 SF)

8 b)

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 2\mathbf{i} + 0\mathbf{j}$ which will always have a non-zero horizontal component, hence the acceleration can never be zero.

9 a)

$\mathbf{s} = \int \mathbf{v}\, dt = (\frac{1}{2}t^2 - 4t + c_1)\mathbf{i} + (\frac{1}{4}t^4 + t^2 + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{s} = \mathbf{0}$

$0 - 0 + c_1 = 0 \Rightarrow c_1 = 0$

$0 + 0 + c_2 = 0 \Rightarrow c_2 = 0$

$\mathbf{s} = (\frac{1}{2}t^2 - 4t)\mathbf{i} + (\frac{1}{4}t^4 + t^2)\mathbf{j}$

9 b)

Solving $\frac{1}{2}t^2 - 4t = 10$ gives $t = -2$ or $t = 10$

$\therefore t = 10$

when $t = 10$

$\mathbf{s} = 10\mathbf{i} + (\frac{1}{4} \times 10^4 + 10^2)\mathbf{j} = 10\mathbf{i} + 2600\mathbf{j}$

2600 metres vertically.

10 a)

$2t^{\frac{1}{2}} = 3t - 1$

$3t - 2t^{\frac{1}{2}} - 1 = 0 \Rightarrow t^{\frac{1}{2}} = 1$ so $t = 1$

(when $t = 1$, $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j}$ which is the same direction)

10 b)

$-2t^{\frac{1}{2}} = 3t - 1$

$3t + 2t^{\frac{1}{2}} - 1 = 0 \Rightarrow t^{\frac{1}{2}} = \frac{1}{3}$ so $t = \frac{1}{9}$

(when $t = \frac{1}{9}$, $\mathbf{v} = \frac{2}{3}\mathbf{i} - \frac{2}{3}\mathbf{j}$ which is the same direction)

10 c)

$13 \times 2t^{\frac{1}{2}} = 3(3t - 1)$

$26t^{\frac{1}{2}} = 9t - 3 \Rightarrow 9t - 26t^{\frac{1}{2}} - 3 = 0$

$\Rightarrow t^{\frac{1}{2}} = 3$ or $t^{\frac{1}{2}} = -\frac{1}{9}$

$t^{\frac{1}{2}} = -\frac{1}{9}$ has no real solutions

$\Rightarrow t = 9$

when $t = 9$, $\mathbf{v} = 6\mathbf{i} + 26\mathbf{j}$ which is the same direction

So $t = 9$ seconds

11 a)

$2t - 36 = 6t^{\frac{1}{2}}$

$\Rightarrow 2t - 6t^{\frac{1}{2}} - 36 = 0$

$\Rightarrow t^{\frac{1}{2}} = 6, t^{\frac{1}{2}} = -3$

So when $t = 36$

11 b)

$\mathbf{r} = \int \mathbf{v}\, dt = (t^2 - 36t + c_1)\mathbf{i} + (4t^{\frac{3}{2}} + c_2)\mathbf{j}$

when $t = 1$, $\mathbf{s} = -20\mathbf{i} + 6\mathbf{j}$

$1 - 36 + c_1 = -20 \Rightarrow c_1 = 15$

$4 + c_2 = 6 \Rightarrow c_2 = 2$

$\mathbf{r} = (t^2 - 36t + 15)\mathbf{i} + (4t^{\frac{3}{2}} + 2)\mathbf{j}$

11 c)

$|\mathbf{v}| = \sqrt{(2t-36)^2 + (6t^{\frac{1}{2}})^2}$

$= \sqrt{4t^2 - 144t + 1296 + 36t}$

$= \sqrt{4t^2 - 108t + 1296}$

$= 2\sqrt{t^2 - 27t + 324}$

So $24 = 2\sqrt{t^2 - 27t + 324}$

$12 = \sqrt{t^2 - 27t + 324}$

$144 = t^2 - 27t + 324$

$t^2 - 27t + 180 = 0$

$t = 12$ or $t = 15$

When $t = 12$

$\mathbf{r} = -273\mathbf{i} + (96\sqrt{3} + 2)\mathbf{j}$

12 a)

$\mathbf{s} = \int \mathbf{v}\, dt = (6t - \frac{2}{3}pt^{\frac{3}{2}} + c_1)\mathbf{i} + (\frac{5}{2}t^2 - qt + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{s} = 4\mathbf{i} + \mathbf{j}$

$0 - 0 + c_1 = 4 \Rightarrow c_1 = 4$

$0 - 0 + c_2 = 1 \Rightarrow c_2 = 1$

$\mathbf{s} = (6t - \frac{2}{3}pt^{\frac{3}{2}} + 4)\mathbf{i} + (\frac{5}{2}t^2 - qt + 1)\mathbf{j}$

when $t = 4$, $\mathbf{s} = 12\mathbf{i} + 21\mathbf{j}$

$6 \times 4 - \frac{2}{3}p \times 4^{\frac{3}{2}} + 4 = 12 \Rightarrow p = 3$

$\frac{5}{2} \times 4^2 - 4q + 1 = 21 \Rightarrow q = 5$

12 b)

$\mathbf{v} = (6 - 3t^{\frac{1}{2}})\mathbf{i} + (5t - 5)\mathbf{j}$

$\mathbf{a} = (\frac{3}{2}t^{-\frac{1}{2}})\mathbf{i} + 5\mathbf{j}$

when $t = 9$, $\mathbf{a} = (\frac{3}{2} \times 9^{-\frac{1}{2}})\mathbf{i} + 5\mathbf{j} = 0.5\mathbf{i} + 5\mathbf{j}$

$|\mathbf{a}| = \sqrt{0.5^2 + 5^2} = \frac{1}{2}\sqrt{101}$

$F = ma$ gives $F = 0.5 \times \frac{1}{2}\sqrt{101} = 2.51$ N (3 SF)

13 a)

$\mathbf{v} = \mathbf{i} + (2kt - 8)\mathbf{j}$

When $t = 0.5$, $2k \times 0.5 - 8 = 0$

$k - 8 = 0 \Rightarrow k = 8$

13 b)

$\mathbf{r} = (t + 1)\mathbf{i} + (8t^2 - 8t)\mathbf{j}$

$|\mathbf{r}| = \sqrt{(t+1)^2 + (8t^2 - 8t)^2}$

$= \sqrt{64t^4 - 128t^3 + 65t^2 + 2t + 1}$

$1 = \sqrt{64t^4 - 128t^3 + 65t^2 + 2t + 1}$

$1 = 64t^4 - 128t^3 + 65t^2 + 2t + 1$

$0 = 64t^4 - 128t^3 + 65t^2 + 2t$

$t = 0$ or $t = -0.0291$

So when $t = 0$ seconds

14 a)

$\mathbf{r}_Q - \mathbf{r}_P = (12 - 2t^2 - 3 + t^2)\mathbf{i} + (4t^2 + 5 - 3t^2 - 4)\mathbf{j}$

$= (9 - t^2)\mathbf{i} + (t^2 + 1)\mathbf{j}$

$d = \sqrt{(9 - t^2)^2 + (t^2 + 1)^2}$

$d^2 = 81 - 18t^2 + t^4 + t^4 + 2t^2 + 1$

$d^2 = 2t^4 - 16t^2 + 82$

14 b)

$\frac{d}{dt}[2t^4 - 16t^2 + 82] = 8t^3 - 32t$

Minimum when $\frac{d}{dt}[2t^4 - 16t^2 + 82] = 0$

$8t^3 - 32t = 0 \Rightarrow 8t(t^2 - 4) = 8t(t - 2)(t + 4) = 0$

So when $t = 0$ and $t = 2$

When $t = 0$, $d^2 = 82$

When $t = 2$, $d^2 = 50$

When $t = 0$ there is a local maximum

Minimised when $t = 2$

15)

$\mathbf{v}_P = \frac{d\mathbf{r}_P}{dt} = (1)\mathbf{i} + (6)\mathbf{j}$

$\mathbf{v}_Q = \frac{d\mathbf{r}_Q}{dt} = (8t)\mathbf{i} + (5)\mathbf{j}$

$|\mathbf{v}_P| = \sqrt{(1)^2 + (6)^2} = \sqrt{37}$

$|\mathbf{v}_Q| = \sqrt{(8t)^2 + (5)^2} = \sqrt{64t^2 + 25}$

$\sqrt{37} = \sqrt{64t^2 + 25}$

$37 = 64t^2 + 25 \Rightarrow 64t^2 = 12 \Rightarrow t^2 = \frac{3}{16}$

$\therefore t = \frac{\sqrt{3}}{4}$

16)

$\mathbf{s}_P = \int \mathbf{v}_P \, dt = (2t^2 + c_1)\mathbf{i} + (\frac{1}{3}t^3 + \frac{2}{3}t + c_2)\mathbf{j}$

$c_1 = 0, c_2 = 0$

$\mathbf{s}_P = (2t^2)\mathbf{i} + (\frac{1}{3}t^3 + \frac{2}{3}t)\mathbf{j}$

when $t = 1$, $\mathbf{s}_P = 2\mathbf{i} + \mathbf{j}$

$|\mathbf{s}_P| = \sqrt{2^2 + 1^2} = \sqrt{5}$

$\mathbf{s}_Q = \int \mathbf{v}_Q \, dt = (t^2 + 7t + c_1)\mathbf{i} + (-4t + c_2)\mathbf{j}$

$c_1 = 0, c_2 = 0$

$\mathbf{s}_Q = (t^2 + 7t)\mathbf{i} + (-4t)\mathbf{j}$

when $t = 1$, $\mathbf{s}_Q = (8)\mathbf{i} + (-4)\mathbf{j}$

$|\mathbf{s}_Q| = \sqrt{8^2 + 4^2} = 4\sqrt{5}$

$4|\mathbf{s}_P| = |\mathbf{s}_Q|$ as req.

17 a)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = 2e^{0.4t}\mathbf{i} + (2t - 4)\mathbf{j}$

17 b)

when $t = 2$, $\mathbf{v} = 2e^{0.48}\mathbf{i}$ so east

17 c)

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 0.8e^{0.4t}\mathbf{i} + 2\mathbf{j}$

when $t = 2$, $\mathbf{a} = 0.8e^{0.8}\mathbf{i} + 2\mathbf{j}$

$|\mathbf{a}| = \sqrt{(0.8e^{0.8})^2 + 2^2}$

$F = m|\mathbf{a}| = 10\sqrt{(0.8e^{0.8})^2 + 2^2} = 26.8$ N

18 a)

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (2t + p)\mathbf{i} + (12t^2)\mathbf{j}$

when $t = 2$, $\mathbf{a} = (4 + p)\mathbf{i} + 48\mathbf{j}$

$|\mathbf{a}| = \sqrt{(4 + p)^2 + 48^2} = \sqrt{16 + 8p + p^2 + 2304}$

$F = ma$ gives $120 = 2 \times \sqrt{p^2 + 8p + 2320}$

$60^2 = p^2 + 8p + 2320 \Rightarrow 3600 = p^2 + 8p + 2320$

$\Rightarrow p^2 + 8p - 1280 = 0$ as req.

18 b)

Solving $p^2 + 8p - 1280 = 0$

$\therefore p = 32$ or $p = -40$

18 c)

No vertical component $\Rightarrow t = 0$

But $t^2 + pt$ would also be 0 so never moves exactly westwards or eastwards.

19)

when $t = 0.1$, $\mathbf{v} = (2 \times 0.1^2)\mathbf{i} + (-5 \times 0.1)\mathbf{j}$

$= 0.02\mathbf{i} - 0.5\mathbf{j}$

$\arctan(\frac{0.5}{0.02}) = 87.70938996\ldots$

$90 + 87.70938996\ldots$

Bearing of 178°

20)

Bearing of 225° $\Rightarrow$ parallel to $-\mathbf{i} - \mathbf{j}$

$-(-t^2 - 4) = -(-t^3)$

$t^2 + 4 = t^3 \Rightarrow t^3 - t^2 - 4 = 0$

$t = 2$

check: $\mathbf{r} = -8\mathbf{i} - 8\mathbf{j}$

$\therefore t = 2$ seconds

21 a)

Bearing of $135°$ $\Rightarrow$ parallel to $\mathbf{i} - \mathbf{j}$

$-(t + \frac{7}{4}) = 2 - t^2$

$-t - \frac{7}{4} = 2 - t^2 \Rightarrow t^2 - t - \frac{15}{4} = 0$

$t = \frac{5}{2}, t = -\frac{3}{2}$

check: $\mathbf{r} = \frac{17}{4}\mathbf{i} - \frac{17}{4}\mathbf{j}$

$\therefore t = 2.5$ seconds

21 b)

$\mathbf{r} = (t + \frac{7}{4})\mathbf{i} + (2 - t^2)\mathbf{j}$

east of origin $\Rightarrow 2 - t^2 = 0 \Rightarrow t = \sqrt{2}$

$\mathbf{v} = \frac{d\mathbf{v}}{dt} = \mathbf{i} + (-2t)\mathbf{j}$

when $t = \sqrt{2}$, $\mathbf{v} = \mathbf{i} + (-2\sqrt{2})\mathbf{j}$

$|\mathbf{v}| = \sqrt{(1)^2 + (-2\sqrt{2})^2} = 3 \text{ ms}^{-1}$

22 a)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (2t)\mathbf{i} + 8\mathbf{j}$

22 b)

$\mathbf{v} = (2T)\mathbf{i} + 8\mathbf{j}$

$2T = 8$ so $T = 4$

$|\mathbf{v}| = \sqrt{8^2 + 8^2} = 11.3 \text{ ms}^{-1}$

23 a)

$-3\mathbf{i} + (t^2 - 3t + 7)\mathbf{j} + (t^2 - 1)\mathbf{i} - 5\mathbf{j} = \mathbf{0}$

$-3 + t^2 - 1 = 0 \Rightarrow t = \pm 2$

$(t^2 - 3t + 7) - 5 = 0 \Rightarrow t^2 - 3t + 2 = 0$

$\Rightarrow t = 1$ or $t = 2$

so $t = 2$

23 b)

$\mathbf{F} = (t^2 - 4)\mathbf{i} + (t^2 - 3t + 2)\mathbf{j}$

$\mathbf{F} = m\mathbf{a}$ so

$(t^2 - 4)\mathbf{i} + (t^2 - 3t + 2)\mathbf{j} = 0.5\mathbf{a}$

$\mathbf{a} = (2t^2 - 8)\mathbf{i} + (2t^2 - 6t + 4)\mathbf{j}$

$\mathbf{v} = \int \mathbf{a}\, dt$

$= (\frac{2}{3}t^3 - 8t + c_1)\mathbf{i} + (\frac{2}{3}t^3 - 3t^2 + 4t + c_2)\mathbf{j}$

when $t = 0$, $v = 0$ so $c_1 = 0$ and $c_2 = 0$

$\mathbf{v} = (\frac{2}{3}t^3 - 8t)\mathbf{i} + (\frac{2}{3}t^3 - 3t^2 + 4t)\mathbf{j}$

North-east:

$\frac{2}{3}t^3 - 8t = \frac{2}{3}t^3 - 3t^2 + 4t$

$0 = -3t^2 + 12t$

$\therefore t = 0$ or $t = 4$

So $t = 4$

23 c)

$\mathbf{v} = (\frac{2}{3} - 8)\mathbf{i} + (\frac{2}{3} - 3 + 4)\mathbf{j}$

$= -\frac{22}{3}\mathbf{i} + \frac{5}{3}\mathbf{j}$

$|\mathbf{v}| = \sqrt{(-\frac{22}{3})^2 + (\frac{5}{3})^2} = \frac{1}{3}\sqrt{509} = 7.52 \text{ ms}^{-1}$

24 a)

$\mathbf{v} = \frac{d\mathbf{s}}{dt} = (2\ln 2 \times 2^{2t})\mathbf{i} - (\ln 2 \times 2^t)\mathbf{j}$

When $t = 0.4$,

$|\mathbf{v}| = \sqrt{(2\ln 2 \times 2^{0.8})^2 + (\ln 2 \times 2^{0.4})^2}$

$= 2.58 \text{ ms}^{-1}$ (3 SF)

24 b)

Bearing of 225° $\Rightarrow$ parallel to $\mathbf{i} - \mathbf{j}$

$-(2^{2t}) = 2^t$

$2^{2t} - 2^t = 0 \Rightarrow 2^t(2^t - 1) = 0 \Rightarrow 2^t - 1 = 0$

$\Rightarrow t = \log_2 1 = 0 \therefore$ when $t = 0$

25 a)

$\mathbf{v} = \int \mathbf{a}\, dt = (8t + c_1)\mathbf{i} + (-2\ln t + c_2)\mathbf{j}$

when $t = 1$, $\mathbf{v} = 3\mathbf{i} - 5\mathbf{j}$

$8 + c_1 = 3 \Rightarrow c_1 = -5$

$0 + c_2 = -5 \Rightarrow c_2 = -5$

$\mathbf{v} = (8t - 5)\mathbf{i} + (-2\ln t - 5)\mathbf{j}$

25 b)

When $t = 0.1$,

$|\mathbf{v}| = \sqrt{(0.8 - 5)^2 + (-2\ln 0.1 - 5)^2}$

$= 4.22 \text{ ms}^{-1}$ (3 SF)

26)

$\mathbf{s} = \int \mathbf{v}\, dt = (e^t + c_1)\mathbf{i} + (-\cos t + c_2)\mathbf{j}$

when $t = 0$, $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$

$1 + c_1 = 2 \Rightarrow c_1 = 1$

$-1 + c_2 = 5 \Rightarrow c_2 = 6$

$\mathbf{s} = (e^t + 1)\mathbf{i} + (-\cos t + 6)\mathbf{j}$

When $t = 0.5$,

$|\mathbf{s}| = \sqrt{(e^{0.5} + 1)^2 + (-\cos 0.5 + 6)^2}$

$= 5.77$ m (3 SF)

27 a)

when $t = 3\pi$

$\mathbf{r} = (4\cos(6\pi))\mathbf{i} + ((3\pi)^2 \sin(3\pi))\mathbf{j} = 4\mathbf{i}$

East of the origin

27 b)

$\mathbf{v} = (-8\sin(2t))\mathbf{i} + (t^2 \cos t + 2t \sin t)\mathbf{j}$

when $t = 3\pi$

$\mathbf{v} = (-8\sin(6\pi))\mathbf{i} + ((3\pi)^2 \cos(3\pi) + 2(3\pi)\sin(3\pi))\mathbf{j}$

$= (-9\pi^2)\mathbf{j}$

South

27 c)

$m = 0.5$

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (-16\cos(2t))\mathbf{i} + (-t^2 \sin t + 2t\cos t + 2t\cos t + 2\sin t)\mathbf{j}$

when $t = 3\pi$

$\mathbf{a} = -16\mathbf{i} - 12\pi\mathbf{j}$

$\mathbf{F} = 0.5(-16\mathbf{i} - 12\pi\mathbf{j}) = -8\mathbf{i} - 6\pi\mathbf{j}$

$|\mathbf{F}| = \sqrt{8^2 + (6\pi)^2} = 20.5$ N (3 SF)

28)

$\int te^{2t} dt = \frac{1}{2}te^{2t} - \frac{1}{2}\int e^{2t} dt = \frac{1}{2}te^{2t} - \frac{1}{4}e^{2t} (+c)$

$\mathbf{v} = \int \mathbf{a}\, dt = (6t + c_1)\mathbf{i} + (\frac{1}{2}te^{2t} - \frac{1}{4}e^{2t} + c_2)\mathbf{j}$

When $t = 0$, $\mathbf{v} = \frac{3}{4}\mathbf{j}$

$0 + c_1 = 0 \Rightarrow c_1 = 0$

$0 - \frac{1}{4} + c_2 = \frac{3}{4} \Rightarrow c_2 = 1$

$\mathbf{v} = (6t)\mathbf{i} + (\frac{1}{2}te^{2t} - \frac{1}{4}e^{2t} + 1)\mathbf{j}$

when $t = 0.2$, $\mathbf{v} = (1.2)\mathbf{i} + (0.1e^{0.4} - \frac{1}{4}e^{0.4} + 1)\mathbf{j}$

$|\mathbf{v}| = \sqrt{1.2^2 + (0.1e^{0.4} - \frac{1}{4}e^{0.4} + 1)^2}$

$= 1.43$ ms^{-1} (3 SF)

29)

$\mathbf{s} = \int_0^{\frac{\pi}{2}} \mathbf{v}\, dt = \int_0^{\frac{\pi}{2}} (\sin 2t)\mathbf{i} + (\cos t)\mathbf{j}\, dt$

$= [(-\frac{1}{2}\cos 2t)\mathbf{i} + (\sin t)\mathbf{j}]_0^{\frac{\pi}{2}}$

$= [(-\frac{1}{2}\cos \pi)\mathbf{i} + (\sin \frac{\pi}{2})\mathbf{j}] - [(-\frac{1}{2}\cos 0)\mathbf{i} + (\sin 0)\mathbf{j}]$

$= (\frac{1}{2}\mathbf{i} + \mathbf{j}) - (-\frac{1}{2}\mathbf{i}) = \mathbf{i} + \mathbf{j}$ metres

$|\mathbf{s}| = \sqrt{1^2 + 1^2} = \sqrt{2} = 1.41$ m (3 SF)

30)

$\mathbf{v} = (-\sin t - e^t)\mathbf{i} + (\cos t - \frac{1}{t})\mathbf{j}$

$= -2.128\ldots\mathbf{i} - 1.122\ldots\mathbf{j}$

$270° - \arctan(\frac{1.122\ldots}{2.128\ldots}) = 242°$

31 a)

$x = 8t - 5, y = 3 + t^2$

$t = \frac{x+5}{8}$ so $y = 3 + (\frac{x+5}{8})^2$

31 b)

$x = t + \frac{7}{4}, y = 2 - t^2$

$t = x - \frac{7}{4}$ so $y = 2 - (x - \frac{7}{4})^2$

31 c)

$x = t^2, y = 5t + 4$

$t = \frac{y-4}{5}$ so $x = (\frac{y-4}{5})^2$

32 a)

$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (\cos t)\mathbf{i} + (-\sin t)\mathbf{j}$

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (-\sin t)\mathbf{i} + (-\cos t)\mathbf{j}$

when $t = \frac{\pi}{2}$

$\mathbf{a} = -\mathbf{i}$

$F = ma = 2 \times 1 = 2$ N

32 b)

$x = \sin t, y = \cos t$

$x^2 + y^2 = (\sin t)^2 + (\cos t)^2 = 1$

so $x^2 + y^2 = 1$